

Forecasting Uncertainty with ML

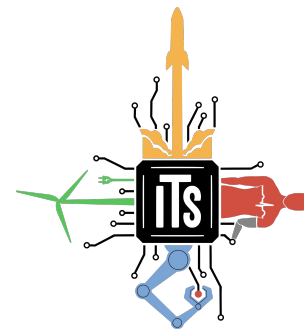


Niklas Erdmann,

PhD student since 2023.

MSc in Artificial Intelligence in Groningen, the Netherlands.

University of Oslo //
The Faculty of Mathematics and Natural Sciences //
Institute of Technology Systems //



Is this enough? - Can we get more than just a forecast?

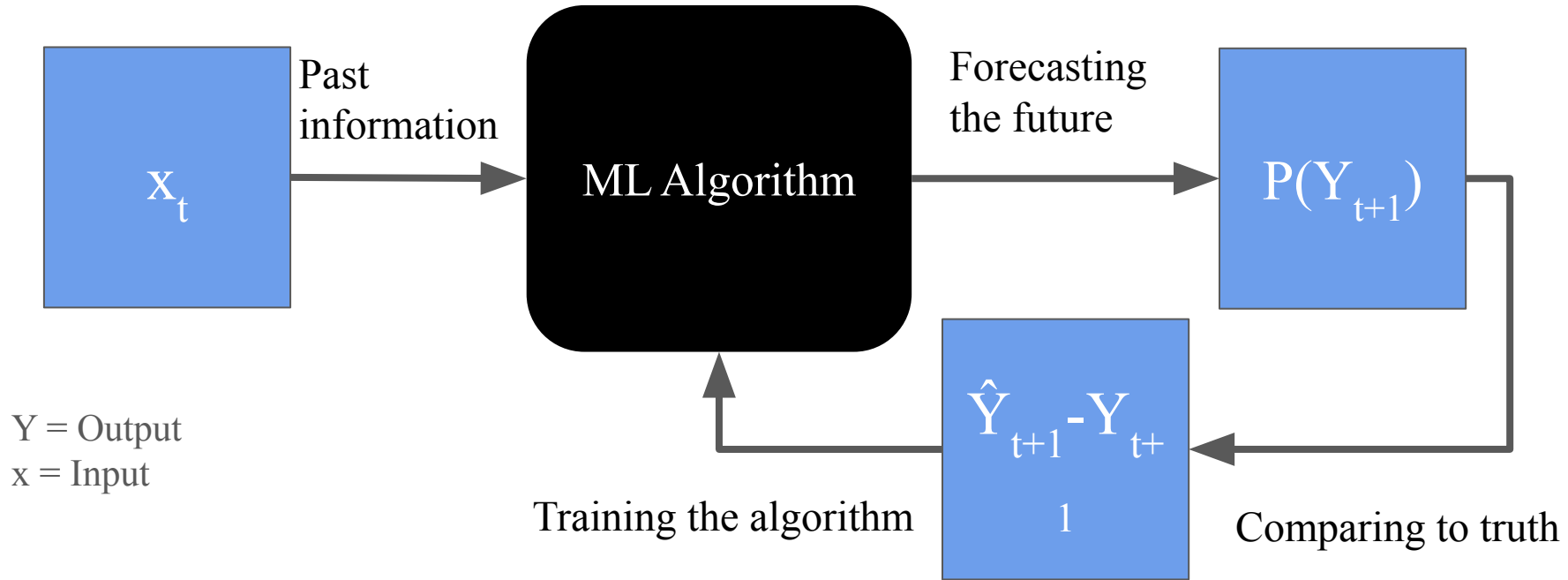
Forecasting can not be perfect.

- We do not know our mistakes directly from a forecast.
- We cannot perfectly predict our mistakes either.

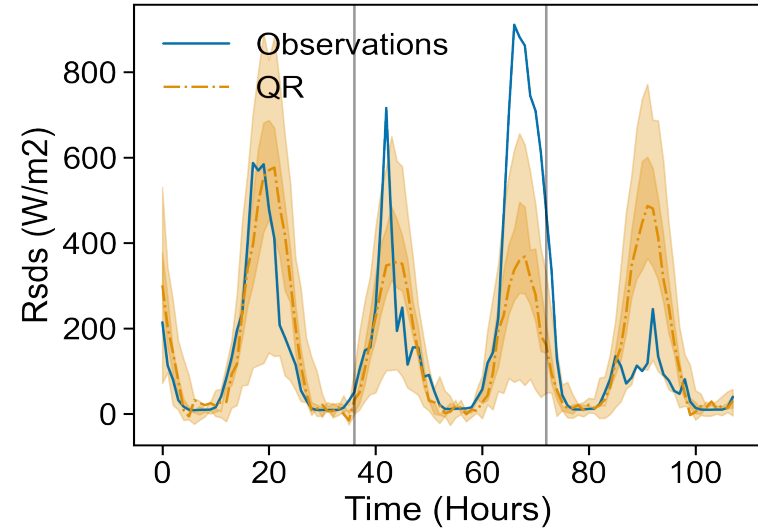
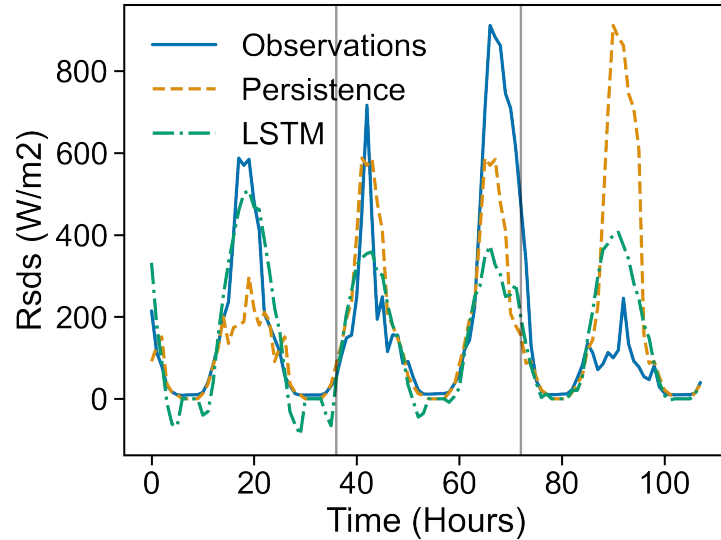
Uncertainty can be forecasted too.

- We can estimate worst case and best case scenarios.
- We can assign probabilities to certain outcomes.
- Richer information makes for better decision making. For humans and machines.

How are we forecasting Probabilities?



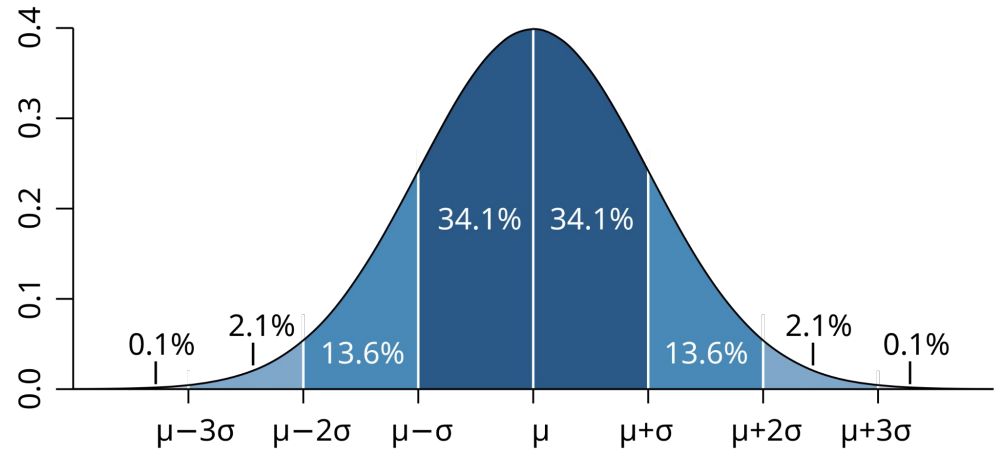
More than just a forecast



N. Erdmann, L. Ø. Bentsen, R. Stenbro, H. N. Riise, N. Warakagoda and P. Engelstad, "Deep and Probabilistic Solar Irradiance Forecast at the Arctic Circle," *2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC)*, Seattle, WA, USA, 2024, pp. 1599-1606, doi: 10.1109/PVSC57443.2024.10748862.

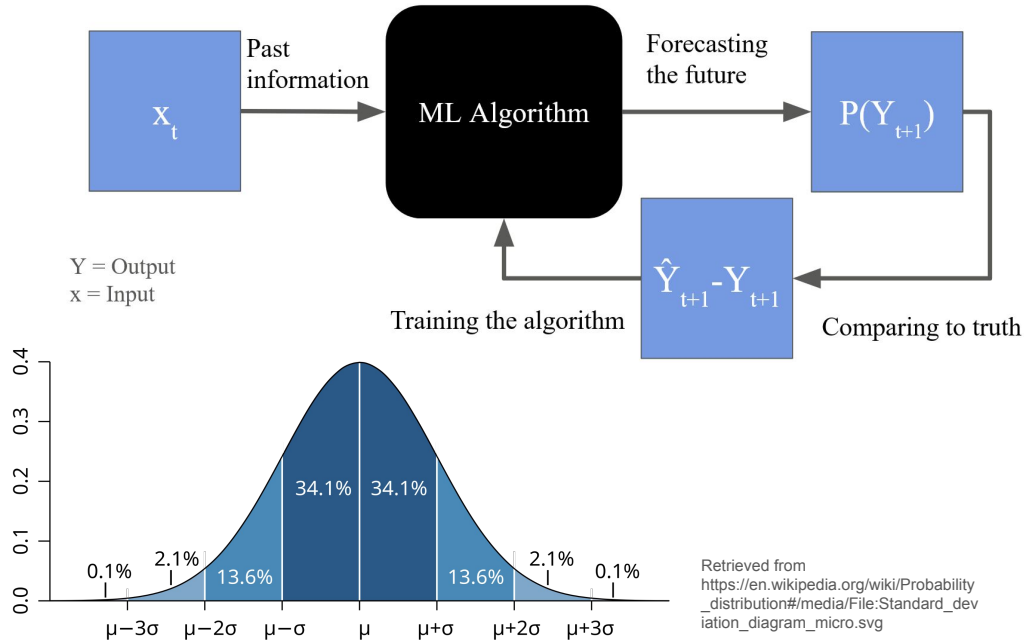
How are we forecasting Probabilities?

1. Change ML- output:
 - a. Probability Distribution
 - b. Quantile
 - c. Distributional sample
2. Change Comparison (loss)
 - a. Maximum Likelihood
 - b. Pinball loss



Retrieved from https://en.wikipedia.org/wiki/Probability_distribution#/media/File:Standard_deviation_diagram_micro.svg

A closer look at Quantile Regression



- Non-continuous and non-parametric
- Limited by output size

New Developments:

- Simultaneous Quantile Regression
- Partially Constrained Machine learning